

# NEO / GICForge Manual

Non-redundant generalized internal coordinates and symmetry-adapted coordinate bases in  
MATRIX

ORACLE Development Notes

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## Abstract

NEO, currently exposed through the compatible **GICForge** command name, is the **ORACLE** geometry backend that converts Cartesian molecular structures into non-redundant generalized internal coordinates (GICs), optional symmetry-adapted GICs, and symmetry-adapted Cartesian coordinates. It is used as a library service by **MORPHEUS**, **SEfit** and future geometry optimizers, and can also be run as a standalone diagnostic tool. This manual describes the input contract, coordinate-generation policy, pruning and symmetrization rules, output files, Python/Fortran identity checks and recommended regression tests.

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## 1 Purpose

NEO/GICForge has one responsibility: starting from a Cartesian molecular structure, it constructs a chemically interpretable, rank-complete coordinate basis. The basis can be used directly as a Gaussian ReadGIC block, frozen as a machine-readable **ORACLE** schema, evaluated analytically on later Cartesian geometries, or converted into symmetry-adapted Cartesian displacements.

**GICForge** deliberately does not own project management, SMILES handling, electronic-structure parsing, DVR, semiexperimental least-squares logic or GUI workflow orchestration. Those tasks belong to the Python layer. The geometry backend owns topology perception from Cartesian coordinates, primitive internal coordinates, ring handling, type-local redundancy removal, symmetry labels and the final auditable coordinate definition.

## 2 Theoretical Background

Let  $x$  be the  $3N$ -dimensional Cartesian coordinate vector and let  $q_p(x)$  be a primitive internal coordinate. **GICForge** first builds a redundant primitive set

$$\mathbf{q}(x) = \{q_1(x), q_2(x), \dots, q_M(x)\},$$

where  $M$  is usually larger than the vibrational rank. A generalized internal coordinate is a linear combination of primitives,

$$Q_k(x) = \sum_p U_{pk} q_p(x),$$

with  $U$  stored in the frozen GIC definition. The Wilson row associated with the coordinate is

$$\mathbf{B}_k = \frac{\partial Q_k}{\partial x} = \sum_p U_{pk} \frac{\partial q_p}{\partial x}.$$

The final coordinate set is valid only if the B matrix has rank  $3N - 6$  for a nonlinear molecule or  $3N - 5$  for a linear molecule.

The central algorithmic choice is that redundancy removal is performed by coordinate family, not by a global decomposition over all candidates. In a global SVD or global rank-revealing decomposition, the numerical ordering of rows can make chemically unrelated coordinates compete with one another. A stretch row could remove a torsional row, or an out-of-plane row could remove a ring bend, even though the two coordinates describe different physical motions. **GICForge** therefore uses the already accepted rows as a rank basis and admits additional rows only within a fixed family order. Inside each family, a modified Gram-Schmidt rank test keeps the rows that add linearly independent B-matrix information. This preserves interpretability while still enforcing the exact vibrational rank.

Symmetry adaptation is a second projection problem. Once the final post-pruning basis is known, equivalent coordinate rows are projected with the molecular symmetry operations and labelled by irreducible representation. The projection is constrained to homogeneous coordinate blocks, so the symmetry operation cannot manufacture a coordinate by mixing unrelated physical families. The theoretical target counts are those of the vibrational representation,

$$\Gamma_{\text{vib}} = \Gamma_{3N} - \Gamma_{\text{trans}} - \Gamma_{\text{rot}}.$$

If the projected coordinate counts cannot reproduce this representation, the coordinate generator fails rather than writing an ambiguous basis.

### 3 Input Contract

The molecular geometry input is the XYZ-format file `xyzin` in the working directory. Atomic symbols and atomic numbers must be equivalent and must cover the full periodic table through the shared `ORACLE` periodic-table utilities. The `provin` file contains only keywords, title, charge and multiplicity; it must not contain Cartesian coordinates, Z-matrices or FCHK-derived geometry.

```
# GNIC BMAT GICSYM

Pyrene GICForge example

0 1
```

Unsupported legacy geometry paths are intentionally disabled: Z-matrix input, Cartesian coordinates embedded in `provin`, FCHK geometry input, SMILES input, and old FIT-POT/VCI/DVR/MSR/isotope/rate driver keywords. Cartesian input is the only production path.

### 4 Build and Execution

The active Fortran backend is built with:

```
cd engines/fortran/gicforge
./compile_legacy
```

The build writes `engines/fortran/gicforge/build/gicforge_legacy`.

The recommended command-line entry point is the shared `ORACLE` CLI:

```
python -m matrix gic --workdir work/gicforge
python -m matrix gic-define geometry.xyz --workdir work/gicdefine
python -m matrix gic-bmatrix oracle.gic.definition.json geometry.xyz
python -m matrix gic-contract --geometry geometry.xyz --workdir work/contract
```

`gic-define` creates a frozen schema; `gic-bmatrix` evaluates that schema without rerunning topology perception; `gic-contract` verifies the Fortran/Python identity contract.

### 5 Coordinate Construction Algorithm

The coordinate generator is a deterministic pipeline. Each step consumes only information created by earlier steps, and the final schema records enough metadata to verify that the same model is reused later.

#### 5.1 Topology perception

`GICForge` starts from interatomic distances and atomic covalent radii. The result is a covalent graph with atoms as vertices and bonds as edges. The graph is checked before coordinate construction: symbols and atomic numbers must be consistent, atom indexes must be unique, and obvious nonbonded H-H contacts or other impossible short contacts are treated as input/topology errors. The graph is then frozen for the run. Later refinement or optimization modules must not allow a geometry step to change this graph silently.

Ring perception is performed on the covalent graph. The ring basis is canonicalized before any ring coordinate is built, so the same molecule gives the same ring order in Fortran and Python.

Fused systems are represented by their individual rings plus shared bonds; the shared bonds are the places where butterfly coordinates can be introduced.

## 5.2 Primitive coordinate generation

The first coordinate layer is primitive and usually redundant. The generator adds primitives in chemically ordered families:

1. stretches for every covalent edge;
2. ordinary valence angles around each bonded center;
3. linear-bend candidates only for nearly linear centers, with a maximum number per center;
4. exocyclic torsions from the default one-dihedral rule;
5. endocyclic valence-angle primitives for each ring;
6. butterfly primitives around bonds shared by two rings;
7. endocyclic puckering components **RPck** for each ring;
8. out-of-plane primitives, except when all four involved atoms are cyclic;
9. optional improper-dihedral primitives in the Gaussian-compatibility mode.

This order is not only cosmetic. It defines the chemical priority used by the rank reduction. Stretches define the bonded frame first; bends define local coordination; torsions and ring coordinates define conformational and puckering directions; out-of-plane coordinates close the remaining local non-planarity directions.

## 5.3 Optional local X–H stretch policy

By default, X–H stretches are ordinary stretches and therefore enter the same symmetry-adaptation path as all other bonded stretches. This is the conservative Merlino-compatible mode and is recorded as **XH\_STRETCH\_POLICY SYMMETRIZE** in the frozen **#GIC** section.

For local-mode anharmonic workflows, selected X–H stretches may instead be kept unsymmetrized. NEO then tags them as a separate **LOCAL\_XH\_STRETCH** family. The coordinates remain analytic **R(i,j)** stretches and participate in rank checking as stretch rows, but the symmetry projector does not combine them into SALCs. This keeps the local oscillators available to GF, VSCF or DVR while leaving the rest of the GIC basis unchanged.

The selection can be made globally with **-xh-stretch-policy local-all**, by one-based bond selectors such as **-local-xh-bond 4-17**, or by local geminal class with **-local-xh-class XH, XH2** or **XH3**. The class is assigned from the number of hydrogens bound to the same heavy atom. **XH2** and **XH3** groups are useful when geminal couplings are expected to be stronger than couplings to the rest of the molecule. If a bond or class selector is given without an explicit policy, the CLI uses **local-selected**. The same fields are written to **#GIC** as **LOCAL\_XH\_BONDS** and **LOCAL\_XH\_CLASSES**, so a later build can reproduce the choice without guessing from labels.

## 5.4 Local combinations before pruning

Several local coordination patterns are better represented by combinations than by a single primitive. For example, equivalent valence angles around a highly symmetric center may be converted to local sums and differences, and ring torsions are converted to deformation coordinates. These combinations are still type-local: angles are combined with angles, torsions with torsions, and out-of-plane coordinates with out-of-plane coordinates. This rule is essential because it prevents the construction stage from creating mathematically valid but chemically opaque coordinates.

For ring systems, endocyclic torsions are not meant to appear as an arbitrary list of raw dihedrals in the final model. They are the primitive source from which **GICForge** builds protected ring-puckering components. **ORACLE** stores the linear **RPck** components in the frozen **GIC** definition because their **B** rows are analytic linear combinations of dihedral **B** rows. The Gaussian block exports the selected **RPck** components directly. The polar  $q, \phi$ -like descriptors are derived in post-processing from those components, because  $\Phi$  is a phase in the puckering subspace rather than a linear symmetry coordinate and is singular at  $Q = 0$ . This rule is used for isolated and fused rings alike.

## 5.5 Rank reduction and final basis

After primitive generation and local combinations, **GICForge** evaluates Wilson **B** rows for the candidate coordinates at the current Cartesian geometry. It then keeps rows in the fixed family order and rejects rows that do not increase the rank. The rank test is numerical, but the candidate competition is chemical: an angle can remove another redundant angle, but not a stretch; a ring bend can be removed by ring-bend redundancy, but not by an unrelated exocyclic torsion.

If the ordinary candidate set is too small before pruning, the backend expands within Fortran to the full primitive candidate set and repeats the same type-local pruning. The number of pre-pruning attempts is capped by the number of available primitive coordinates. If the final count is still not  $3N - 6$  or  $3N - 5$ , the run stops. A downstream solver must never repair the basis by inventing an independent coordinate model.

## 6 Coordinate Definition Format

Coordinate definitions are printed in a Gaussian ReadGIC-compatible form. Each line has a name on the left-hand side, an equal sign, and either one primitive or a normalized linear combination of primitives on the right-hand side:

```
Str0001 = R(1,2)
Bend0007 = A(2,1,6)
Tors0014 = D(3,4,5,6)
OuPl0003 = U(7,2,1,3)
GIC0021 = 0.70710678 A(2,1,6) -0.70710678 A(3,4,5)
```

Atom indexes are one-based and always refer to the input atom order after any explicit canonical ring ordering has been recorded in the report. The primitive symbols are:

| Primitive    | Meaning                                                            |
|--------------|--------------------------------------------------------------------|
| $R(i, j)$    | Bond stretch between atoms $i$ and $j$ .                           |
| $A(i, j, k)$ | Valence angle $i - j - k$ , with $j$ as the central atom.          |
| $L(i, j, k)$ | Linear-bend component for a nearly linear $i - j - k$ arrangement. |

| Primitive       | Meaning                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------|
| $D(i,j,k,l)$    | Dihedral angle around the central bond $j - k$ .                                                                             |
| $U(i,j,k,l)$    | Out-of-plane coordinate with a single central atom. The Fortran and Python parsers enforce the same central-atom convention. |
| $ImpD(i,j,k,l)$ | Improper-dihedral compatibility form used only with <code>G16</code> or <code>ImpDih</code> .                                |

The name prefix indicates the coordinate family and is part of the stable ordering contract:

| Prefix       | Family                                                                                         |
|--------------|------------------------------------------------------------------------------------------------|
| <b>Str</b>   | primitive stretch;                                                                             |
| <b>Bend</b>  | exocyclic or final valence-angle coordinate;                                                   |
| <b>LinB</b>  | linear-bend coordinate;                                                                        |
| <b>Tors</b>  | exocyclic torsion;                                                                             |
| <b>BtFl</b>  | butterfly torsion around a bond shared by two rings;                                           |
| <b>CyGNA</b> | endocyclic valence-angle combination;                                                          |
| <b>QPck</b>  | post-processing ring-puckering amplitude-like descriptor built from <b>RPck</b> components;    |
| <b>PhiP</b>  | post-processing ring-puckering phase-like descriptor built from <b>RPck</b> components;        |
| <b>RPck</b>  | frozen ORACLE linear ring-puckering component with analytic B row and Gaussian ReadGIC export; |
| <b>OuPl</b>  | out-of-plane coordinate;                                                                       |
| <b>ImpD</b>  | improper-dihedral compatibility coordinate.                                                    |

## 6.1 Ring Coordinate Format

For a ring with ordered atoms  $a_1, a_2, \dots, a_n$ , `GICForge` generates two ring families. Endocyclic valence-angle coordinates are combinations of primitive angles

$$A(a_{i-1}, a_i, a_{i+1}),$$

where indexes are cyclic. Endocyclic torsional coordinates are combinations of primitive dihedrals

$$D(a_{i-1}, a_i, a_{i+1}, a_{i+2}).$$

The primitive torsions are first transformed to **RPck** components. The frozen ORACLE B matrix and Gaussian ReadGIC export use these linear components directly. Consecutive selected **RPck** pairs can be converted after the run to puckering descriptors:

$$Q = \sqrt{RPck_i^2 + RPck_j^2}, \quad \Phi = \text{atan2}(RPck_j, RPck_i).$$

For a five-membered ring, the report therefore separates the coordinate construction from the active coordinates:

```
Endocyclic Dihedral Angles
5-Membered Ring ( 13 12 16 1 2): 2 Dihedral angles

Final ring torsion coordinates
```

```

RPck0001 = ...
RPck0002 = ...

Post-processing descriptors
  QPck0001 = ...
  PhiP0001 = ...

```

For six-membered and larger isolated rings, the same rule is used: the primitive basis is endocyclic torsional, the optimized Gaussian coordinates are the selected **RPck** components, and the  $q, \phi$ -like descriptors are computed later. Fused rings are treated consistently: each elementary fused-ring cycle contributes its own protected ring-puckering source space, the fused set is reduced only within the protected ring family, and the surviving **RPck** components are exported directly. Fused rings also add **BtF1** butterfly coordinates around shared bonds. Butterfly coordinates remain in the torsional family for pruning because their B rows describe ring-puckering motion.

For direct Gaussian **ReadAllGIC** optimizations, the exported block must remain a complete non-redundant internal-coordinate basis. Although **GICForge** records the total-symmetric irrep and the list of total-symmetric GICs in the frozen **xyzin** contract, Gaussian’s Berny optimizer still requires the full internal basis to build its metric. The Gaussian input writer therefore exports all final GICs, but marks every non-totally-symmetric final coordinate as **Frozen**. Only the totally symmetric block remains variable in a symmetry-preserving Gaussian optimization. MATRIX-controlled optimizers and least-squares workflows use the saved total-symmetric list to select symmetry blocks internally without relying on Gaussian syntax.

## 7 Coordinate Families

**GICForge** builds primitive coordinates from the saved covalent graph unless a fragment mode requests otherwise. Weak complexes have two supported models. In **SPECIAL\_COORDINATES** mode, fragments remain separate bodies and their relative motion is described by protected fragment-center, fragment-translation and fragment-orientation coordinates. In **PSEUDO\_BONDS** mode, **GICForge** first tries the Merlino/BDPCS3 X–H $\cdots$ Y hydrogen-bond detector; if no chemically valid hydrogen bond connects the relevant fragments, it falls back to the closest inter-fragment atom pair. The selected H $\cdots$ Y or closest contact is added only to the GIC construction graph. It is serialized in **xyzin** under **[PSEUDO\_BONDS]** and does not alter the saved covalent topology. Pseudo-bonds do not generate artificial rings: ring coordinates are built only from the original covalent ring section.

| Family            | Policy                                                                                                                                                                                                                                           |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stretch           | All bonded pairs are kept as primitive <b>R(i,j)</b> coordinates. Stretches are not reduced by the final pruning step.                                                                                                                           |
| Local X–H stretch | Optional unsymmetrized <b>R(X,H)</b> rows selected by <b>-xh-stretch-policy</b> , <b>-local-xh-bond</b> or <b>-local-xh-class</b> . They remain stretch coordinates but form their own symmetry block and are kept local for VSCF/DVR workflows. |
| Exocyclic bend    | Valence angles outside rings are generated around each center and reduced locally by rank.                                                                                                                                                       |
| Linear bend       | Linear-angle candidates are generated only when required by local geometry and are capped per center to avoid unphysical proliferation.                                                                                                          |

| Family            | Policy                                                                                                                                                                                                                                                                                                                     |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exocyclic torsion | The default torsion generator is the one-dihedral <b>ONEDIH</b> path. The keyword <b>NOONEDIH</b> forces the older alternative for diagnostics.                                                                                                                                                                            |
| Ring bend         | Endocyclic valence-angle combinations are generated ring by ring, with canonical ring numbering.                                                                                                                                                                                                                           |
| Ring torsion      | Endocyclic torsions are converted to <b>RPck</b> components before pruning. The frozen <b>ORACLE</b> basis keeps selected <b>RPck</b> rows for linear analytic B-matrix evaluation and Gaussian export. <b>QPck</b> and <b>PhiP</b> are derived post-processing descriptors, not active Gaussian optimization coordinates. |
| Butterfly         | Fused-ring butterfly coordinates are generated around shared bonds and kept as torsional ring-family coordinates.                                                                                                                                                                                                          |
| Out-of-plane      | Default non-planarity primitives are <b>U(...)</b> out-of-plane angles. They are not generated when all four involved atoms are cyclic. Cyclic centers with exocyclic substituents remain eligible.                                                                                                                        |
| Improper dihedral | With <b>G16</b> or <b>ImpDih</b> , improper dihedrals are written instead of out-of-plane angles. This is a compatibility mode and must match the Python parser exactly.                                                                                                                                                   |

## 8 Rings and Polycyclic Systems

Ring atom ordering is canonicalized with the same Prelog-first convention used by the Python implementation. The **provout** report lists atoms in one cycle, atoms linking multiple cycles, rings found versus expected rings, endocyclic valence-angle coordinates, endocyclic torsional coordinates, butterfly coordinates and out-of-plane candidates.

For planar polycyclic aromatic hydrocarbons, out-of-plane candidates whose four atoms are all cyclic are suppressed. Exocyclic out-of-plane coordinates remain available and are either kept or removed by the final rank test. Regression targets should include naphthalene, anthracene, pyrene and coronene, with explicit checks that fused-ring butterfly coordinates are present when topologically required.

## 9 Redundancy Removal

The final pruning step is type-local. **GICForge** builds B rows for candidate GICs and applies a modified Gram–Schmidt rank test separately to homogeneous coordinate families:

- exocyclic bends;
- linear bends;
- exocyclic torsions;
- butterfly and ring torsional coordinates;
- ring valence-angle coordinates;
- out-of-plane coordinates.

Stretch rows are always retained as primitive coordinates. This policy avoids the chemically undesirable behavior of a global SVD in which, for example, a stretch can eliminate a torsion simply because it appears earlier in the global rank basis. The rank test is still numerical and B-matrix based, but the admissible competition is restricted to coordinates with the same physical role.

Fragment and virtual-center coordinates are treated as a separate protected class. These coordinates include fragment-center distances, fragment-center to atom distances, fragment translations, fragment orientations and atom to virtual-center distances for ring centers, bond centers and contact centers. They are not ordinary valence-graph primitives, and they can carry chemical information that no single stretch, bend or torsion should be allowed to erase. For this reason **GICForge** first tests all protected special coordinates against each other by the same analytic B-matrix modified Gram-Schmidt rank test used by the Fortran77 routine **ORCGSEL**. Only after that protected pass does it use ordinary primitives to complete the vibrational rank. A special coordinate may be skipped only when its own row is singular or linearly dependent on previously accepted special rows. If the independent protected set alone is larger than the vibrational rank, **GICForge** stops with a contract error instead of silently deleting a protected coordinate.

For a non-covalent complex this policy gives two deliberately different coordinate bases for the same geometry. The fragment model keeps inter-fragment rigid-body coordinates protected. The pseudo-bond model removes fragment coordinates from the selected basis and lets the standard stretch/bend/torsion/OOP generator operate on the graph augmented by the selected weak contact. Primitive candidates are still reduced by analytic B-matrix rank; the pseudo-bond is an internal-coordinate edge, not a request to keep every primitive generated around it.

The frozen **xyzin** schema records this distinction explicitly. Ordinary coordinates are written with **CLASS=ORDINARY**. Fragment and virtual-center coordinates are written with the protected class token **SPECIAL\_PROTECTED**. The header also records the rank method **analytic\_b\_matrix\_mgs\_greedy**. The reduction policy is the protected-first policy described above, serialized in **xyzin** as two adjacent tokens: **SPECIAL\_PROTECTED\_FIRST\_THEN\_ORDINARY\_** and **ANALYTIC\_RANK**. This makes restarts reproducible and tells downstream programs that the protected coordinates are part of the molecular model, not disposable redundant candidates.

The target rank is  $3N - 6$  for nonlinear molecules and  $3N - 5$  for linear molecules. If the final non-redundant count does not match the target, **GICForge** stops with an error. If the pre-pruning count differs from the target, coordinate definitions are not printed as final coordinates until after pruning has produced the usable basis.

## 10 Symmetry

**GICForge** uses symmetry in three different places, and these roles should not be confused.

### 10.1 Point-group perception

The first role is molecular point-group perception. **GICForge** links the shared Fortran symmetry engine **symm.f**. Cartesian coordinates are centered at the center of mass, oriented on principal inertia axes and assigned a point group. The authoritative report is the point group computed by **symm.f**; an optional point-group token on the second **xyzin** line is legacy metadata only.

The symmetry report records the detected point group, the quality of the assignment and the maximum atom-matching deviation. A high-symmetry assignment is used only when atom permutations and Cartesian operations are consistent within the configured tolerances. If the geometry is only approximately symmetric, the report must make that explicit, because the symmetry-adapted coordinate basis is only as meaningful as the atom mapping used to build it.

## 10.2 Symmetry labels for final GICs

The second role is assigning irreducible representations to the final post-pruning GIC basis. This is done after the non-redundant coordinate set has been selected. The order matters: symmetry is not allowed to rescue a redundant coordinate set, and pruning is not allowed to destroy the final irreducible-representation counts. The target representation is

$$\Gamma_{\text{vib}} = \Gamma_{3N} - \Gamma_{\text{trans}} - \Gamma_{\text{rot}}.$$

For a nonlinear molecule, the sum of the target irrep counts is therefore  $3N - 6$ .

With `GICSYM`, the deterministic Python post-check projects the final GIC rows onto symmetry-adapted combinations, assigns irreducible representations, writes the symmetry-adapted GIC block and records the result in:

- `gauin.symm`;
- `gicsym`;
- `gic_symmetry_diagnostics.json`.

The promoted `gauin` file contains the symmetrized Gaussian block. Totally symmetric coordinates are written first, followed by a blank separator and then the other irreducible representations. Downstream programs must not re-symmetrize; they must consume the frozen schema and its labels.

## 10.3 Allowed symmetry blocks

The post-check is deliberately constrained. A symmetry-adapted coordinate must be expressible from a physically homogeneous source block. The allowed blocks are primitive stretches, ordinary bends, linear bends, torsions, out-of-plane coordinates, ring bends, ring torsions, local out-of-plane/improper blocks around the same center and protected special-coordinate blocks. Protected special-coordinate blocks are fragment-center distances, fragment-center to atom distances, fragment translations, fragment orientations and atom to virtual-center distances. There is no global fallback that mixes every primitive in the molecule. This preserves the meaning of the final coordinates and makes the result comparable between Fortran and Python.

Pseudo-bond H-bond coordinates are not placed in protected special blocks. Once the pseudo-bond is selected, the resulting stretches, bends and torsions belong to the ordinary homogeneous valence blocks. This mirrors the Fortran `HBOND/MkHBnd` idea: connect the graph through a weak  $\text{H}\cdots\text{Y}$  contact, then use ordinary internal coordinates.

Within a homogeneous block, equivalent primitives are collected by the symmetry operations. The totally symmetric combination is the normalized sum over the orbit. Non-totally-symmetric combinations are orthonormal differences or the corresponding projection-operator rows. The sign is canonicalized so repeated runs produce the same labels and coefficients. The post-check then verifies both irrep counts and coordinate-family counts. For example, a missing torsional  $B_{2g}$  coordinate cannot be replaced by an extra stretch of the same irrep.

Signed coordinates carry the parity of the Cartesian operation. Torsions, improper dihedrals, out-of-plane coordinates, butterfly coordinates and `RPck` ring-puckering components are pseudoscalar rows: after the atom permutation and canonical coordinate orientation have been applied, their primitive transformation is multiplied by  $\det R$ . This prevents planar out-of-plane or puckering

coordinates from being labelled as totally symmetric under a molecular-plane reflection. Stretches, valence angles and ring bends remain scalar coordinates and do not receive this determinant factor. Ring-puckering components require one additional invariant because **RPck** coordinates are already linear combinations of redundant endocyclic dihedrals. For **RPck** source spaces of any dimension, **MATRIX** can verify the projector in the physical B-row source space used by the analytic Wilson B matrix. This is a coordinate-family rule, not a molecule-specific patch: the symmetry operation must map the selected puckering-source rows into the same row space that downstream GF, fitting and optimization modules will differentiate. When the Merlino-compatible primitive-space projector already closes the block, the B-row projector is accepted only if it preserves the same irrep sequence. If the primitive projector does not close but a finite independent B-row source representation does, the B-row representation can be used to avoid an artificial local-SALC fallback. If neither representation closes, **MATRIX** stops or records the explicit fallback instead of inventing a new coordinate family.

## 10.4 Fortran local SALCs versus production symmetry

The keyword **SYMMALL** activates a limited Fortran-side local SALC pass for selected non-stretch one-term coordinates. This is a construction aid, not the owner of production symmetry. Production symmetry adaptation is the **GICSYM** post-check, because it sees the final pruned basis, can compare against the theoretical vibrational representation and writes the frozen schema. Stretches remain primitive in the raw **gauin** and are symmetry-adapted only in the post-check when **GICSYM** is requested.

## 10.5 Symmetry-adapted Cartesians

With **SYCART**, **GICForge** writes symmetry-adapted Cartesian coordinates after removal of rotations and translations. This provides a common service for **SEfit**, **MORPHEUS** and future optimizers without embedding symmetry logic in those modules. The Cartesian basis follows the same point-group assignment as the **GIC** basis. It is useful when a downstream module wants a translationally and rotationally clean totally symmetric displacement space without using **GICs**.

## 10.6 Failure policy for symmetry

Symmetry adaptation fails if the projected basis cannot reproduce the target vibrational irrep counts, if coordinate-family counts change, if atom permutations are inconsistent, or if two equivalent source blocks would produce ambiguous labels. These failures are intentional. A wrong or ambiguous symmetry label would contaminate downstream refinement, Gaussian optimization or force-field projection more seriously than a clean stop.

A second, independent validation is performed by GF/PED when **-symmetry-blocks** is requested. In a correctly symmetrized **GIC** basis, the internal *F* and *G* matrices cannot contain couplings between different irreducible representations, except for numerical roundoff. **MATRIX** therefore treats such off-block couplings as a contract error to be fixed in the coordinate/symmetry layer; it does not silently solve the full problem and label the result as symmetry-blocked.

# 11 Frozen Schema and Public API

The production public API is the **GICForge** service in **oracle\_gic**. A fresh definition run writes a **oracle.gic.definition.v1** schema containing:

- primitive coordinate expressions;
- GIC coefficient matrix;
- coordinate labels and names;
- coordinate families;
- irreducible representations;
- point group and symmetrization flag;
- Gaussian-readable GIC block;
- provenance hashes for input, backend and generated files.

The schema is frozen during an ordinary refinement or optimization. Later iterations evaluate values and B rows from the schema without changing topology, ring detection, redundancy removal or symmetry assignment. A restart is the explicit boundary at which the schema may be regenerated.

## 12 Output Files

| File                                       | Meaning                                                                                                                                                                                               |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>provout</code>                       | Human-readable report: point group, topology summary, candidate families, pruning diagnostics, final counts and final coordinate definitions.                                                         |
| <code>gicforge_report.txt</code>           | MATRIX-native report generated from frozen <code>xyzin</code> state with rank method, reduction diagnostics, family counts, protected-coordinate counts, symmetry source blocks and final GIC labels. |
| <code>gauin</code>                         | Gaussian ReadGIC input block for the final post-pruning coordinate basis. With <code>GICSYM</code> , this is the promoted symmetrized block.                                                          |
| <code>gauin.raw</code>                     | Preserved unsymmetrized Gaussian block when <code>GICSYM</code> promotes <code>gauin.symm</code> .                                                                                                    |
| <code>gauin.symm</code>                    | Symmetry-adapted Gaussian block before promotion.                                                                                                                                                     |
| <code>gicsym</code>                        | Names, irreps and source-block information for the symmetrized coordinates.                                                                                                                           |
| <code>bmat.out</code>                      | Machine-readable final B matrix when <code>BMAT</code> is active.                                                                                                                                     |
| <code>bondout</code>                       | Detected covalent topology.                                                                                                                                                                           |
| <code>ringin</code>                        | Canonical ring data used by the ring-coordinate generator.                                                                                                                                            |
| <code>sycart.xyz</code>                    | Symmetry-adapted Cartesian coordinates when <code>SYCARD</code> is active.                                                                                                                            |
| <code>oracle.gic.definition.v1.json</code> | Frozen reusable GIC schema written by <code>gic-define</code> .                                                                                                                                       |

## 13 Keyword Summary

| Keyword     | Effect                                                                                                                                                                                                                                                     |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GNIC        | Build generalized natural/internal coordinates.                                                                                                                                                                                                            |
| BMAT        | Write the machine-readable final B matrix.                                                                                                                                                                                                                 |
| GICSYM      | Produce symmetry-adapted GICs and promote them to the Gaussian input block.                                                                                                                                                                                |
| SYCART      | Produce symmetry-adapted Cartesian coordinates without rotations and translations.                                                                                                                                                                         |
| GDV         | Use out-of-plane $U(\dots)$ coordinates; this is also the default behavior.                                                                                                                                                                                |
| G16, ImpDih | Use Gaussian-style improper dihedrals instead of out-of-plane coordinates.                                                                                                                                                                                 |
| ONEDIH      | One-dihedral torsion generation. This is the default.                                                                                                                                                                                                      |
| NOONEDIH    | Force the legacy alternative torsion path for diagnostics.                                                                                                                                                                                                 |
| SYMMALL     | Apply the limited type-local Fortran SALC generator before final pruning. Production symmetry adaptation is still owned by GICSYM.                                                                                                                         |
| HBOND       | Legacy Merlino keyword for detecting $X-H\cdots Y$ contacts. In MATRIX/NEO the corresponding mode is <code>-fragment-mode pseudo-bonds</code> ; detected H-bonds are preferred over the closest-pair fallback and are written as <code>KIND=HBOND</code> . |
| JOINFR      | Legacy Merlino fragment-joining concept. MATRIX keeps the saved covalent topology unchanged and records any weak graph-joining contact in <code>#GIC/[PSEUDO_BONDS]</code> .                                                                               |

The Python CLI exposes the fragment choice explicitly:

```
matrix neo plan molecule.xyzin --fragment-mode special-coordinates
matrix neo build molecule.xyzin --fragment-mode special-coordinates

matrix neo plan molecule.xyzin --fragment-mode pseudo-bonds
matrix neo build molecule.xyzin --fragment-mode pseudo-bonds
```

For the formic-acid-water probe used in the regression suite, the first mode produces protected fragment distance/translation/orientation coordinates with full B rank, while the second mode detects the  $O-H\cdots O$  contact, writes `1 5 6 KIND=HBOND`, and produces only ordinary stretch, bend and torsion coordinates.

## 14 Python/Fortran Identity Contract

The Python layer is allowed to orchestrate, serialize, post-check symmetry and evaluate the frozen B matrix. It is not allowed to replace a valid Fortran GIC basis silently. The contract checks:

- final coordinate count and family counts;
- ordered primitive signatures;
- point group and irreducible-representation labels;
- number of totally symmetric coordinates;

- absence of cross-irrep  $F/G$  couplings in symmetry-block GF/PED diagnostics;
- Python analytic B matrix against Fortran `bmat.out`;
- byte-stable symmetry diagnostics for repeated runs.

Run the contract with:

```
python -m matrix gic-contract --geometry examples/pyrene.xyz \
--workdir work/gic_contract_pyrene
```

The imported Merlino3.0 strict Fortran77 backend is vendored and compiled with:

```
engines/fortran/gicforge/legacy_merlino
engines/fortran/gicforge/compile_legacy
```

ORACLE-specific Fortran fragment/TRIC kernels remain in:

```
engines/fortran/gicforge/frag_tric_bmat.f
```

Both layers must mirror `oracle_gicforge.policy`: primitive families, reduction classes, rank method, protected-first pruning and symmetry source blocks are shared contract data, not backend-local choices.

The readable ORACLE report is generated directly from the frozen `xyzin`:

```
matrix gicforge report molecule.xyzin gicforge_report.txt
```

The Python port in `oracle_gic.gicforge_python` is a faithful porting target, not an independent production generator. It must match Fortran on ordering, primitive orientation, coordinate families, type-local pruning, symmetry labels and B rows before it can replace a backend path.

## 15 Recommended Regression Set

The minimum regression set should cover:

- acyclic molecules with ordinary stretches, bends and torsions;
- linear-angle cases with capped linear bends;
- square-planar or high-coordination centers;
- SF<sub>6</sub>, which should not produce spurious linear bends;
- saccharine and other heteroatom rings;
- naphthalene, anthracene, pyrene and coronene for fused PAHs;
- norbornene and norbornadiene for bridged non-planar rings;
- spiro systems with two rings sharing one atom, which must keep the Merlino **Spir** inter-ring angle block;
- GICSYM variants for nontrivial point groups;
- SYCART variants for symmetry-adapted Cartesian output.

Every test should verify the target rank, coordinate-family counts, presence of required butterfly coordinates, absence of all-cyclic out-of-plane primitives, and equality of Python/Fortran B matrices where both paths are active.

## 16 Failure Policy

**GICForge** must fail early when the generated basis is not scientifically usable. Fatal conditions include:

- inconsistent symbol/atomic-number input;
- topology changes during a frozen-schema workflow;
- final GIC count different from  $3N - 6$  or  $3N - 5$ ;
- symmetry post-check unable to reach theoretical irrep counts;
- coordinate-family counts changed by post-check symmetrization;
- Python/Fortran B-matrix mismatch beyond numerical roundoff.

These failures are intentional. A downstream fit or optimizer must not proceed with an ambiguous coordinate definition.

## 17 Relation to MORPHEUS and SEfit

MORPHEUS and SEfit call **GICForge** as a coordinate-service library. A fresh fit generates one frozen coordinate definition at the starting geometry. Ordinary least-squares iterations then reuse that definition and only update coordinate values, B rows and projected parameter directions. This prevents the fitted model from changing during optimization and makes the final refinement auditable.

For a GIC active model, the fit uses the totally symmetric block assigned by **GICForge**. For a symmetry-adapted Cartesian active model, the fit uses **SYCART**. In both cases, topology and symmetry belong to **GICForge**, while weights, constraints, predicates, parameter classes and least-squares diagnostics belong to the refinement layer.